

Problem 2

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Problem 1 Suppose that $r(t) = r_0 e^{-kt}$ (with $k > 0$) is the rate at which a nation extracts oil. The current rate of extraction is $r(0) = 10^7$ barrels/yr. Also assume that the estimate of the total oil reserve (ie, the amount of oil remaining beneath the ground in this country) is 2×10^9 barrels.

- (a) Find $A(t)$, the total amount of oil extracted by the nation after t years.

Explanation. $A(t) = \int_0^t r(s) ds$

$$\begin{aligned} A(t) &= \int_0^t r(s) ds \\ &= \int_0^t r_0 e^{-ks} ds \\ &= -\frac{r_0}{k} \left[e^{-ks} \right]_0^t \\ &= -\frac{r_0}{k} (e^{-kt} - 1) \\ &= -\frac{1}{k} 10^7 (e^{-kt} - 1) \end{aligned}$$

- (b) Evaluate $\lim_{t \rightarrow \infty} A(t)$ and explain the meaning of this limit.

Explanation.

$$\begin{aligned} \lim_{t \rightarrow \infty} A(t) &= \lim_{t \rightarrow \infty} -\frac{1}{k} 10^7 (e^{-kt} - 1) \\ &= -\frac{1}{k} 10^7 (0 - 1) \\ &= \frac{1}{k} 10^7. \end{aligned}$$

- (c) Find the minimum constant k for which the total oil reserves will last forever.

Explanation. For the oil reserves to last forever, we need that

$$\begin{aligned}\lim_{t \rightarrow \infty} A(t) &\leq 2 \times 10^9 \\ \iff \frac{1}{k} 10^7 &\leq 2 \times 10^9 \\ \iff \frac{1}{k} &\leq 2 \times 10^2 = 200 \\ \iff \frac{1}{200} &\leq k.\end{aligned}$$

So the minimum value for k is $\frac{1}{200}$.

- (d) Suppose that the constant k is half the minimum value found in part (c). When will the nation deplete its oil reserve?

Explanation. First note that $k = \frac{1}{2} \cdot \frac{1}{200} = \frac{1}{400}$. We want to find the value of t such that:

$$\begin{aligned}A(t) &= 2 \times 10^9 \\ -400 \times 10^7 \left(e^{-\frac{1}{400}t} - 1 \right) &= 2 \times 10^9 \\ \left(e^{-\frac{1}{400}t} - 1 \right) &= \frac{2 \times 10^9}{-400 \times 10^7} = -\frac{1}{2} \\ e^{-\frac{1}{400}t} &= \frac{1}{2} \\ -\frac{1}{400}t &= \ln \left(\frac{1}{2} \right) = -\ln(2) \\ t &= 400 \ln(2) \approx 277.259 \text{ years.}\end{aligned}$$